

OUJDA ANGADS, Morocco

WMO: 601150

Lat: 34.787N

Lon: 1.924W

Elev: 1535

StdP: 13.90

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	33.8	36.1	23.1	18.4	52.8	26.9	22.0	52.7	33.9	60.3	29.9	57.9	4.1	240	0.482

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	25.9	100.7	69.1	96.9	69.2	93.6	68.9	75.1	90.3	73.7	88.1	72.3	85.6	13.0	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.1	121.3	80.1	69.6	115.3	79.5	68.2	109.6	78.3	39.7	90.6	38.2	88.2	36.9	85.5	85.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.8	22.2	19.9	30.1	108.9	2.5	3.4	28.3	111.4	26.8	113.4	25.4	115.3	23.5	117.8
			28.5	78.4	2.7	2.0	26.6	79.8	25.0	81.0	23.5	82.1	21.5	83.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	64.5	51.2	52.4	56.3	60.0	66.3	73.8	79.9	80.5	73.8	67.8	58.2	53.0	(6)
	DBStd	11.47	5.05	5.07	5.02	5.00	5.63	5.16	4.81	4.39	4.69	5.35	5.84	5.10	(7)
	HDD50	115	46	30	8	0	0	0	0	0	0	0	4	27	(8)
	HDD65	1894	428	354	272	165	50	1	0	0	2	31	217	374	(9)
	CDD50	5412	83	96	204	301	506	714	926	947	713	552	251	119	(10)
	CDD65	1718	0	1	4	16	91	265	461	482	265	118	14	1	(11)
	CDH74	18273	3	13	109	282	1061	2844	5304	5277	2220	1028	116	16	(12)
	CDH80	8860	0	3	25	71	381	1314	2942	2867	901	332	22	2	(13)
(14) Wind	WSAvg	8.2	8.7	8.2	8.1	8.4	8.4	8.6	8.7	8.4	7.7	7.3	8.1	8.2	(14)
(15) Precipitation	PrecAvg	13.2	1.1	1.4	1.7	1.8	1.2	0.5	0.1	0.2	0.6	1.1	1.3	1.5	(15)
	PrecMax	21.3	4.1	5.1	5.5	7.2	6.5	3.0	0.5	1.1	3.5	6.1	4.6	7.1	(16)
	PrecMin	4.3	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
	PrecStd	4.1	1.0	1.3	1.4	1.6	1.4	0.7	0.1	0.3	0.8	1.4	1.0	1.8	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	73.2	75.1	83.7	87.6	95.1	100.6	106.5	106.1	98.2	93.3	83.7	75.2	(19)
		MCWB	53.9	55.6	59.6	62.2	64.0	68.4	69.7	68.7	69.0	64.4	60.0	55.7	(20)
	2%	DB	68.4	70.0	76.7	81.2	88.0	95.2	100.7	100.6	91.8	87.7	77.1	69.8	(21)
		MCWB	53.0	54.2	57.5	61.1	63.9	69.1	69.4	69.3	67.9	63.2	57.9	54.7	(22)
	5%	DB	64.7	66.6	72.5	76.8	84.0	91.4	97.0	96.7	88.2	83.6	73.1	66.3	(23)
		MCWB	52.4	53.8	56.9	59.5	63.4	67.9	69.3	69.9	67.6	63.9	57.8	53.9	(24)
	10%	DB	61.9	63.4	68.4	72.9	80.1	87.6	93.5	93.2	85.4	80.1	69.6	63.0	(25)
		MCWB	51.6	52.9	56.2	58.4	62.4	66.9	69.0	69.7	67.3	63.5	57.0	52.7	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	57.6	59.0	63.0	66.6	69.7	74.7	76.5	77.4	74.5	70.2	65.1	59.5	(27)
		MCDB	65.9	68.7	74.9	80.0	85.4	91.5	93.6	92.8	86.6	82.4	72.9	67.2	(28)
	2%	WB	55.5	57.0	60.5	63.4	67.2	72.3	74.7	75.5	72.2	68.0	62.2	57.2	(29)
		MCDB	64.2	65.6	71.6	75.6	81.3	88.4	91.9	89.5	83.6	78.4	71.0	65.6	(30)
	5%	WB	53.9	55.4	58.7	61.3	65.3	70.4	73.2	74.1	70.7	66.3	60.2	55.4	(31)
		MCDB	62.4	64.1	69.1	72.8	79.0	85.6	88.9	87.5	81.5	77.3	68.8	63.7	(32)
	10%	WB	52.2	53.7	57.0	59.3	63.5	68.4	71.6	72.7	69.4	64.9	58.2	53.8	(33)
		MCDB	60.2	61.6	66.2	69.5	76.3	82.7	85.9	85.3	79.8	75.4	66.6	61.5	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	20.2	21.1	22.8	23.7	24.3	25.5	26.5	25.9	23.5	23.1	20.5	20.1	(35)
		MCDBR	25.3	26.9	30.9	31.7	31.8	31.0	31.9	30.6	27.9	28.6	25.3	24.3	(36)
	5% WB	MCWBR	14.2	15.1	16.2	15.9	14.0	13.0	12.7	12.2	11.9	12.3	12.2	13.2	(37)
		MCWBR	21.6	23.8	26.2	26.9	26.6	27.3	26.9	25.3	23.1	24.2	21.8	21.2	(38)
(40) Clear-Sky Solar Irradiance	taub		0.299	0.326	0.383	0.409	0.444	0.476	0.526	0.499	0.451	0.395	0.343	0.310	(40)
		taud	2.525	2.450	2.278	2.211	2.136	2.044	1.911	1.994	2.134	2.295	2.442	2.518	(41)
	Ebn at Noon		290	292	281	278	269	259	245	249	256	262	269	278	(42)
		Edh at Noon	26	31	40	45	49	54	61	55	46	36	28	25	(43)
(44) All-Sky Solar Radiation	RadAvg		921	1205	1596	1936	2207	2442	2406	2170	1763	1345	974	841	(44)
	RadStd		81	99	104	136	151	75	58	66	78	85	80	59	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only			+0.86	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)			+0.40	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page